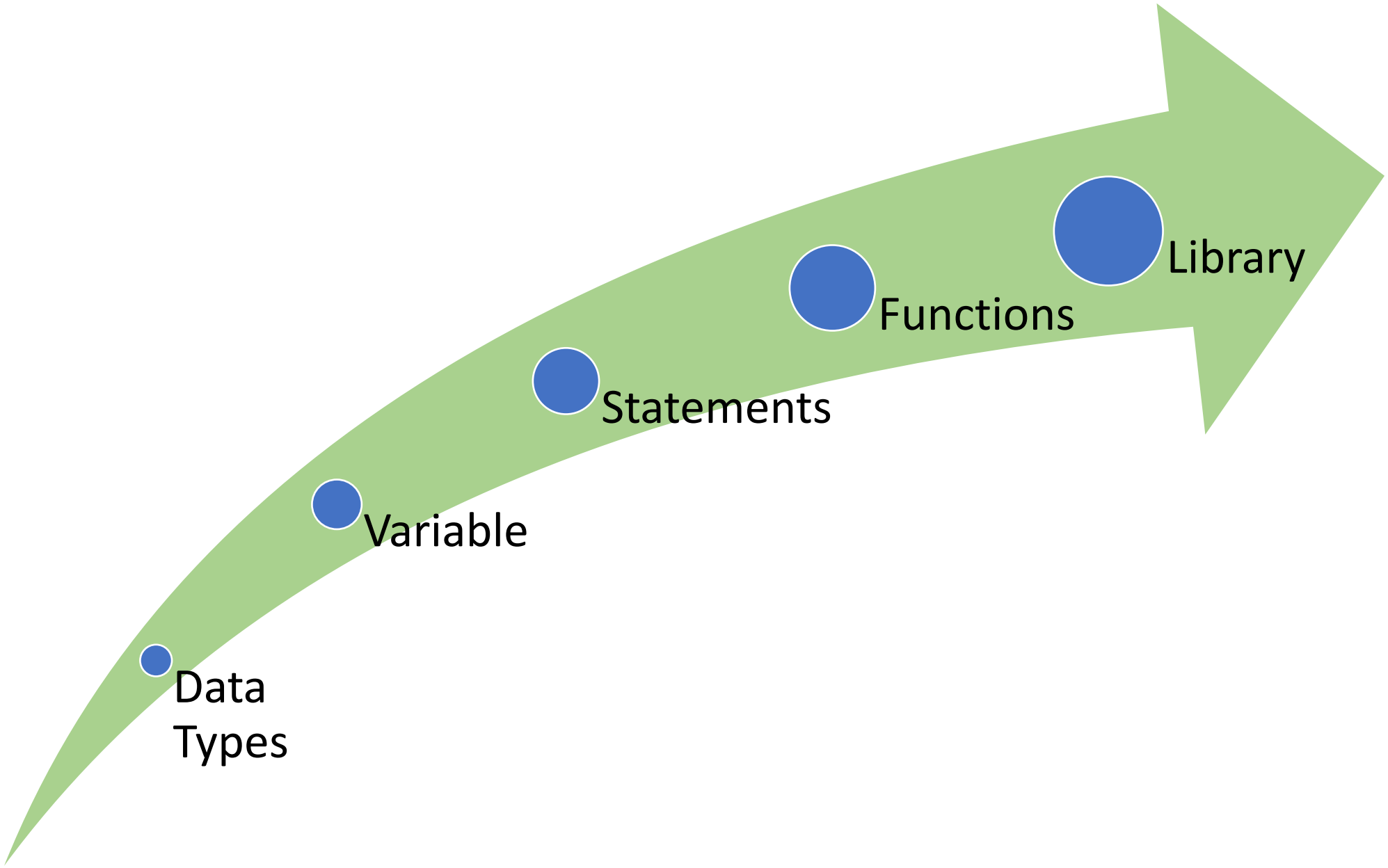
CSC3002
C/C++ Programming

Functions and Libraries

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Data
Types
Variable
Statements
Functions
Library

Functions and Libraries

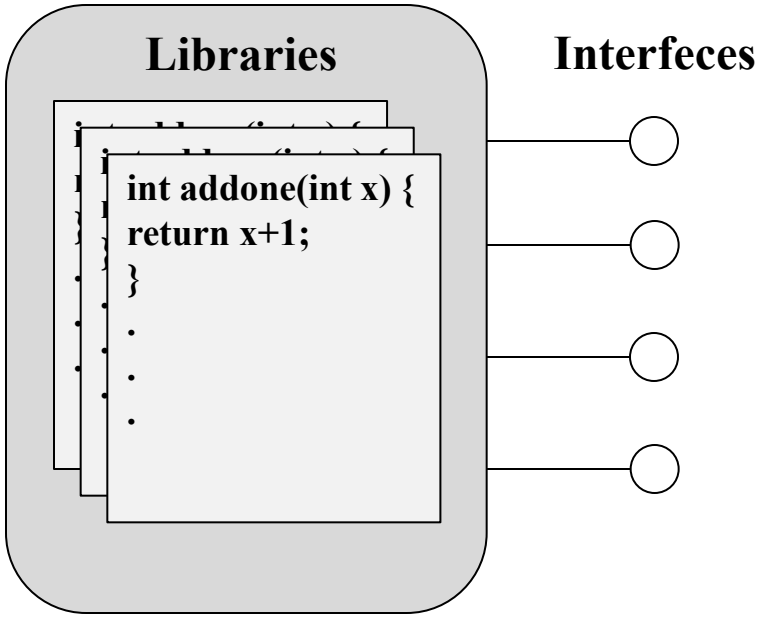
2.1 The idea of a function
2.2 Defining functions in C++
2.3 The mechanics of function calls
2.4 Reference parameters
2.5 Operator
2.6 Libraries
2.7 Introduction to the C++ libraries
2.8 Interfaces and implementations
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Libraries

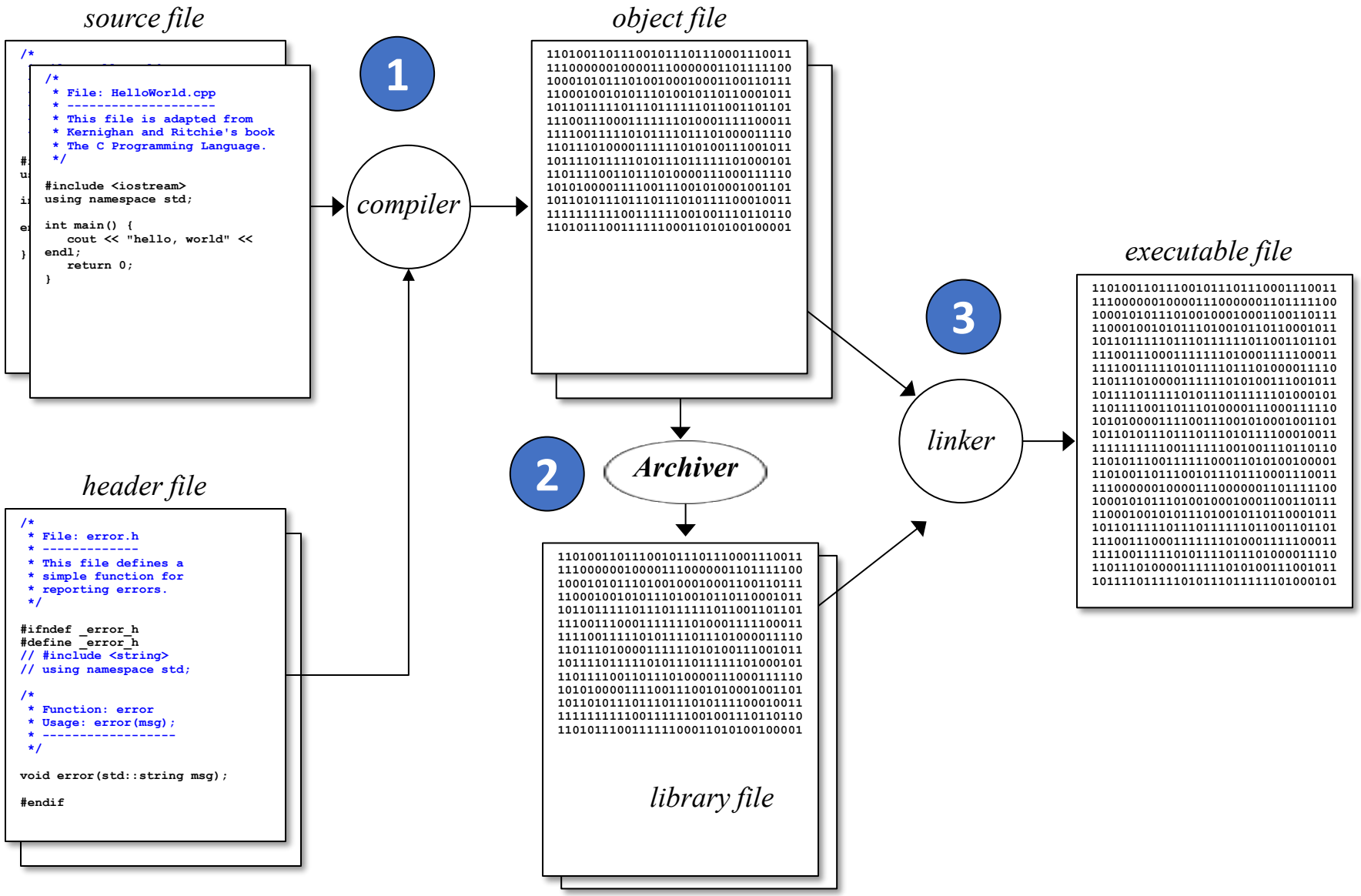
- Modern programming depends on the use of libraries. When you create a typical application, you write only a tiny fraction of the code.
- In computer science, a library is a collection of non-volatile resources used by computer programs, often to develop software.
- A library is also a collection of implementations of behavior, written in terms of a language, that has a well-defined interface by which the behavior is invoked.
- If you want to become an effective programmer, you need to spend at least as much time learning about the libraries as you do learning the language itself.

Libraries

- Libraries can be viewed from two perspectives. Code that uses a library is called a client. The code for the library itself is called the implementation.
- The point at which the client and the implementation meet is called the interface, which serves as both a barrier and a communication channel:
  - Sharing and reuse
  - Abstraction and hiding



The Compilation Process



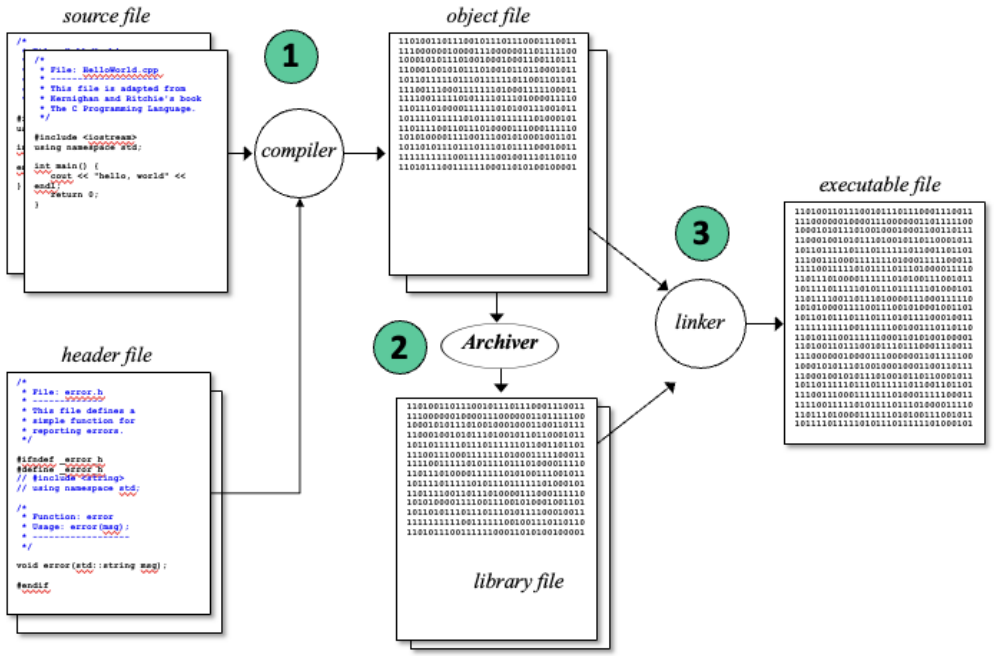
Commands for creating and using a LIB

1. #crerate a object file.
   g++ -c direction.cpp -o direction.o

   …

2. #generate the lib, r-replace, c-create, s-symbol table)
   ar rcs dir.lib direction.o

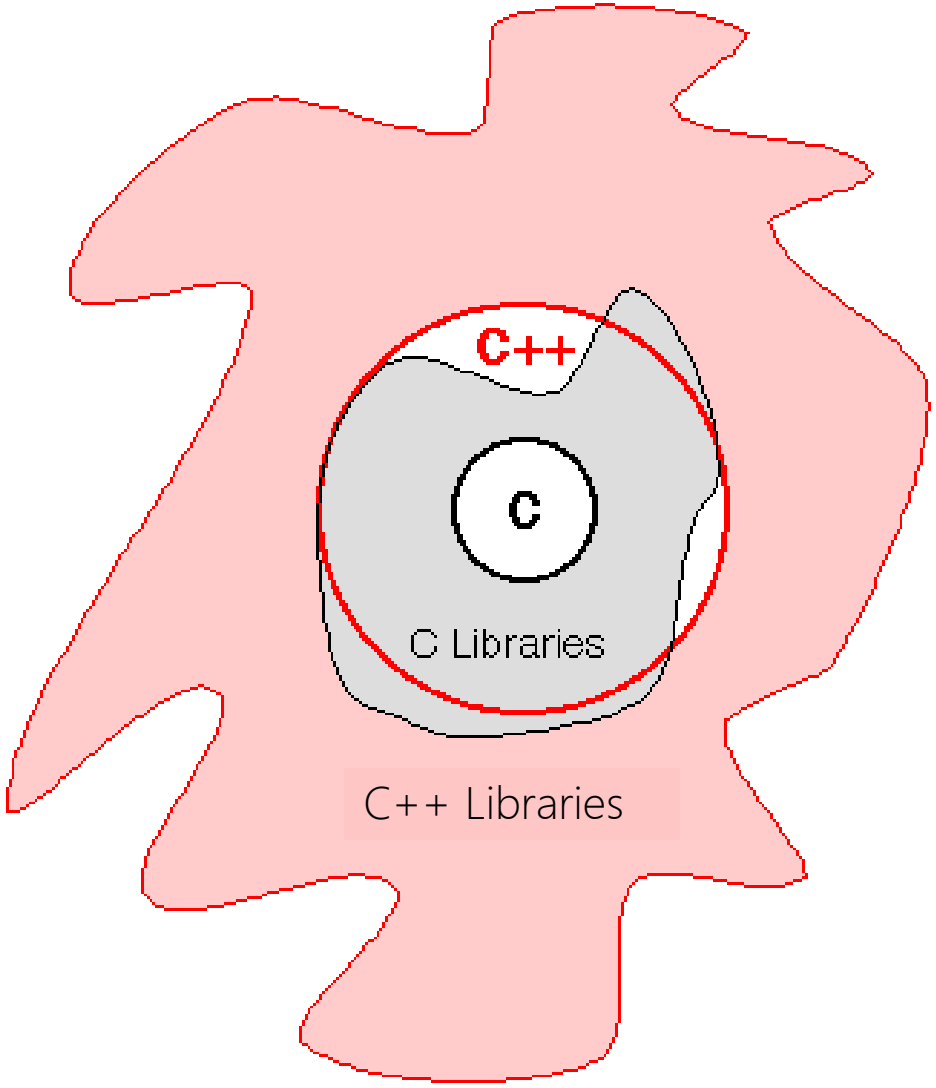
3. #create App using lib
   g++ DirDemo.cpp -L. dir.lib -o main



Relationship Between C and C++

- One of the fundamental design principles of C++ was that it would contain C as a subset, which is also part of the reason behind the success of C++. This design strategy makes it possible to convert applications from C to C++ incrementally.

- The downside of this strategy is that the design of C++ is less consistent and integrated than it might otherwise be.



Introduction to the C++ Standard Libraries

- A collection of classes and functions, which are written in the core language and part of the C++ ISO Standard itself. Features of the C++ Standard Library are declared within the std namespace
  - Containers: vector, queue, stack, map, set, etc.
  - General: algorithm, functional, iterator, memory, etc.
  - Strings
  - Streams and Input/Output: iostream, fstream, sstream, etc.
  - Localization
  - Language support
  - Thread support library
  - Numerics library
  - C standard library: cmath, cctype, cstring, cstdio, cstdlib, etc.

Useful Functions in the <cmath> Library

abs(x)	Returns the absolute value of x.
sqrt(x)	Returns the square root of x.
floor(x)	Returns the largest integer less than or equal to x.
ceil(x)	Returns the smallest integer greater than or equal to x.
exp(x)	Returns the exponential function of x (e^x).
log(x)	Returns the natural logarithm (base e) of x.
log10(x)	Returns the common logarithm (base 10) of x.
pow(x, y)	Returns x^y.
cos(theta)	Returns the trigonometric cosine of the radian angle theta.
sin(theta)	Returns the sine of the radian angle theta.
tan(theta)	Returns the tangent of the radian angle theta.
atan(x)	Returns the principal arctangent of x.
atan2(y, x)	Returns the arctangent of y divided by x.

http://www.cplusplus.com/reference/. http://www.cppreference.com/

From Client to Implementer

- Libraries can be viewed from two perspectives. Code that uses a library is called a client. The code for the library itself is called the implementation.
- Consequently, a programmer who calls functions provided by a library is also called a client of that library. A programmer who implements a library is called an implementer.
- As we go through the chapters in this book, you will have a chance to look at several libraries (e.g., the collection classes) from both of these perspectives, first as a client and later as an implementer.
- We will now switch from a client's perspective to an implementer's.

Creating Library Interfaces

In C++, libraries are made available to clients through the use of an interface file that has the suffix .h, which designates a header file, as illustrated by the following error.h file:

```
/*
 * File: error.h
 * -------------
 * This file defines a simple function for reporting errors.
 */

#ifndef _error_h
#define _error_h
 #include <string>
// using namespace std;

/*
 * Function: error
 * Usage: error(msg);
 * ------------------
 * Writes the string msg to the cerr stream and then exits the program
 * with a standard status code indicating failure.
 */

void error(std::string msg);

#endif
```

Interfaces require standardized preprocessor directive definitions (called boilerplate) to ensure that the interface file is read only once during a compilation.

`using namespace` *should generally NOT be used in a header file, because it will probably be included by many other source files. Therefore all references to standard libraries in header files must include the* `std::` *marker.*

Implementing Library Interfaces

In C++, the header file contains only the prototypes of the exported functions. The implementations of those functions appear in the corresponding .cpp file:

```
/*
 * File: error.cpp
 * ---------------
 * This file implements the error.h interface.
 */

#include <iostream>        // cerr, endl
#include <cstdlib>         // exit, EXIT_FAILURE
#include <string>          // string
#include "error.h"
using namespace std;

/*
 * This function writes out the error message to the cerr stream and
 * then exits the program. The EXIT_FAILURE constant is defined in
 * <cstdlib> to represent a standard failure code.
 */

void error(string msg) {
   cerr << msg << endl;
   exit(EXIT_FAILURE);
}
```

Implementation files typically include their own interface. That is also why error.h does not include **<string>** *but still works.*

using namespace *should always come after the inclusion of all the libraries and header files, because it affects all the code after it.*

Exporting types

Let's create an interface that exports one of the enumerated types, such as the Direction type used to encode the four standard compass points:

```
/*
 * File: direction.h
 * -----------------
 * This interface exports an enumerated type called Direction whose
 * elements are the four compass points: NORTH, EAST, SOUTH, and WEST.
 */

#ifndef _direction_h
#define _direction_h

/*
 * Type: Direction
 * ---------------
 * This enumerated type is used to represent the four compass directions.
 */

enum Direction { NORTH, EAST, SOUTH, WEST };
```

```
/*
 * Function: leftFrom
 * Usage: Direction newdir = leftFrom(dir);
 * ----------------------------------------
 * Returns the direction that is to the left of the argument.
 * For example, leftFrom(NORTH) returns WEST.
 */

Direction leftFrom(Direction dir);

/*
 * Function: rightFrom
 * Usage: Direction newdir = rightFrom(dir);
 * -----------------------------------------
 * Returns the direction that is to the right of the argument.
 * For example, rightFrom(NORTH) returns EAST.
 */

Direction rightFrom(Direction dir);

/*
 * Function: directionToString
 * Usage: string str = directionToString(dir);
 * -------------------------------------------
 * Returns the name of the direction as a string.
 */

std::string directionToString(Direction dir);
```

```
/*
 * Operator: <<
 * Usage: cout << dir;
 * -------------------
 * Overloads the << operator so that it is able to display Direction values.
 */

std::ostream & operator<<(std::ostream & os, Direction dir);

/*
 * Operator: ++
 * Usage: ++dir
 * ------------
 * Overloads the prefix version of the ++ operator to work with Direction
 * values.
 */

Direction operator++(Direction & dir);

/*
 * Operator: ++
 * Usage: dir++
 * ------------
 * Overloads the suffix version of the ++ operator to work with Direction
 * values, to support, e.g., the idiom
 *
 *    for (Direction dir = NORTH; dir <= WEST; dir++) . . .
 */

Direction operator++(Direction & dir, int);

#endif
```

Exporting types

- Implementation of the Direction library

```
/*
 * File: direction.cpp
 * -------------------
 * This file implements the direction.h interface.
 */

#include <string>
#include "direction.h"
using namespace std;

/*
 * Implementation notes: leftFrom, rightFrom
 * -----------------------------------------
 * These functions use the remainder operator to cycle through the
 * internal values of the enumeration type.  Note that the leftFrom
 * function cannot subtract 1 from the direction because the result
 * might then be negative; adding 3 achieves the same effect but
 * ensures that the values remain positive.
 */

Direction leftFrom(Direction dir) {
   return Direction((dir + 3) % 4);
}
Direction rightFrom(Direction dir) {
   return Direction((dir + 1) % 4);
}
```

```
/*
 * Implementation notes: directionToString
 * ---------------------------------------
 * Most C++ compilers require the default clause to make sure that this
 * function always returns a string, even if the direction is not one
 * of the legal values.
 */

std::string directionToString(Direction dir) {
   switch (dir) {
    case NORTH: return "NORTH";
    case EAST: return "EAST";
    case SOUTH: return "SOUTH";
    case WEST: return "WEST";
    default: return "???";
   }
}

/*
 * Implementation notes: <<
 * ------------------------
 * This operator must return the stream by reference after printing
 * the value.  The operator << returns this stream, so the function
 * can be implemented as a single line.
 */

std::ostream & operator<<(std::ostream & os, Direction dir) {
   return os << directionToString(dir);
}
```

```
/*
 * Implementation notes: ++
 * ------------------------
 * The int parameter in the signature for this operator is a marker used
 * by the C++ compiler to identify the suffix form of the operator.  Note
 * that the value after incrementing a variable containing WEST will be
 * out of the Direction range.  That fact will not cause a problem if
 * this operator is used only in the for loop idiom for which it is defined.
 */

Direction operator++(Direction & dir) {
   dir = Direction(dir + 1);
   return dir;
}

Direction operator++(Direction & dir, int) {
   Direction old = dir;
   dir = Direction(dir + 1);
   return old;
}
```

If you want **dir** *to go back to* **NORTH** *after* **WEST**, *how should you modify the code?*

Exporting Constants

```
/*
 * File: gmath.h
 * -------------
 * This file exports the constant PI along with a few degree-based
 * trigonometric functions, which are typically easier to use.
 */

#ifndef _gmath_h
#define _gmath_h

/* Constants */

extern const double PI;                    /* stant pi */

/*
 * Function: sinDegrees
 * Usage: double sine = sinDegrees(angle);
 * ---------------------------------------
 * Returns the trigonometric sine of angle expressed in degrees.
 */
```

In C++, constants written in this form are private to the source file that contains them and cannot be exported through an interface. To export the constant **PI**, *you need to add the keyword* **extern** *to both its definition and the prototype declaration in the interface.*

```
double sinDegrees(double angle);

/*
 * Function: cosDegrees
 * Usage: double cosine = cosDegrees(angle);
 * -----------------------------------------
 * Returns the trigonometric cosine of angle expressed in degrees.
 */

double cosDegrees(double angle);

/*
 * Function: toDegrees
 * Usage: double degrees = toDegrees(radians);
 * -------------------------------------------
 * Converts an angle from radians to degrees.
 */

double toDegrees(double radians);

/*
 * Function: toRadians
 * Usage: double radians = toRadians(degrees);
 * -------------------------------------------
 * Converts an angle from degrees to radians.
 */

double toRadians(double degrees);

#endif
```

Exporting Constants

```
/*
 * File: gmath.cpp
 * ---------------
 * This file implements the gmath.h interface.  In all cases, the
 * implementation for each function requires only one line of code,
 * which makes detailed documentation unnecessary.
 */

#include <cmath>
#include "gmath.h"

extern const double PI = 3.14159265358979323846;

double sinDegrees(double angle) {
   return sin(toRadians(angle));
}

double cosDegrees(double angle) {
   return cos(toRadians(angle));
}

double toDegrees(double radians) {
   return radians * 180 / PI;
}

double toRadians(double degrees) {
   return degrees * PI / 180;
}
```

In C++, constants written in this form are private to the source file that contains them and cannot be exported through an interface. To export the constant **PI**, *you need to add the keyword* **extern** *to both its definition and the prototype declaration in the interface.*

Principles of Interface Design

- **Unified**. Every library should define a consistent abstraction with a clear unifying theme. If a function does not fit within that theme, it should not be part of the interface.
- **Simple**. The interface design should simplify things for the client. To the extent that the underlying implementation is itself complex, the interface must seek to hide that complexity.
- **Sufficient**. For clients to adopt a library, it must provide functions that meet their needs. If some critical operation is missing, clients may decide to abandon it and develop their own tools.
- **General**. A well-designed library should be general enough to meet the needs of many different clients. A library that offers narrowly defined operations for one client is not nearly as useful as one that can be used in many different situations.
- **Stable**. The functions defined in a class exported by a library should maintain precisely the same structure and effect, even as the library evolves. Making changes in the behavior of a library forces clients to change their programs, which reduces its utility.



Designing a Random Number Library

- Nondeterministic behavior turns out to be difficult to achieve on a computer. A computer executes its instructions in a precise, predictable way. If you give a computer program the same inputs, it will generate the same outputs every time, which is not what you want in a nondeterministic program.
- Given that true nondeterminism is so difficult to achieve in a computer, libraries such as the random.h interface described in this chapter must instead simulate randomness by carrying out a deterministic process that satisfies the following criteria:
  1. The values generated by that process should be difficult for human observers to predict.
  2. Those values should appear to be random, in the sense that they should pass statistical tests for randomness.
- Because the process is not truly random, the values generated by the random.h interface are said to be pseudorandom.



Designing a Random Number Library

- The function rand from <cstdlib> takes no arguments and returns an integer between 0 and RAND_MAX randomly. It is not sufficient for the clients.
- Choosing the right set of functions
  - ✓ Selecting a random integer in a specified range
  - ✓ Choosing a random real number in a specified range
  - ✓ Simulating a random event with a specific probability
- The designers of the C++ library (and the earlier C libraries) decided that rand should return the same random sequence each time a program is run, so that it is possible to use rand in a deterministic way in order to support debugging.
  - ✓ Set the initial value (seed) of the random number generating process, e.g., the time when the program is run



The random.h Interface

```
/*
 * File: random.h
 * --------------
 * This file exports functions for generating pseudorandom numbers.
 */

#ifndef _random_h
#define _random_h

/*
 * Function: randomInteger
 * Usage: int n = randomInteger(low, high);
 * ----------------------------------------
 * Returns a random integer in the range low to high, inclusive.
 */

int randomInteger(int low, int high);
```



The random.h Interface

```
/*
 * Function: randomReal
 * Usage: double d = randomReal(low, high);
 * ----------------------------------------
 * Returns a random number in the half-open interval [low, high).
 * A half-open interval includes the first endpoint but not the
 * second.
 */

double randomReal(double low, double high);

/*
 * Function: randomChance
 * Usage: if (randomChance(p)) ...
 * -------------------------------
 * Returns true with the probability indicated by p.  The
 * argument p must be a floating-point number between 0 (never)
 * and 1 (always).
 */

bool randomChance(double p);
```



The random.h Interface

```
/*
 * Function: randomBool
 * Usage: if (randomBool()) ...
 * ----------------------------
 * Returns <code>true</code> with 50% probability.
 */
bool randomBool();

/*
 * Function: setRandomSeed
 * Usage: setRandomSeed(seed);
 * ---------------------------
 * Sets the internal random number seed to the specified value.
 * You can use this function to set a specific starting point
 * for the pseudorandom sequence or to ensure that program
 * behavior is repeatable during the debugging phase.
 */

void setRandomSeed(int seed);

#endif
```



The **random.cpp** Implementation

```
/*
 * File: random.cpp
 * ----------------
 * This file implements the random.h interface.
 */

#include <cstdlib>
#include <cmath>
#include <ctime>
#include "random.h"
using namespace std;

/* Private function prototype */

void initRandomSeed();
```



The random.cpp Implementation

```
/*
 * Implementation notes: randomInteger
 * -----------------------------------
 * The code for randomInteger produces the number in four steps:
 *
 * 1. Generate a random real number d in the range [0 .. 1).
 * 2. Scale the number to the range [0 .. N).
 * 3. Translate the number so that the range starts at low.
 * 4. Truncate the result to the next lower integer.
 *
 * The implementation is complicated by the fact that both the
 * expression RAND_MAX + 1 and the expression high - low + 1 can
 * overflow the integer range.
 */

int randomInteger(int low, int high) {
   initRandomSeed();
   double d = rand() / (double(RAND_MAX) + 1);
   double s = d * (double(high) - low + 1);
   return int(floor(low + s));
}
```



The random.cpp Implementation

```
/*
 * Implementation notes: randomReal
 * --------------------------------
 * The code for randomReal is similar to that for randomInteger,
 * without the final conversion step.
 */

double randomReal(double low, double high) {
   initRandomSeed();
   double d = rand() / (double(RAND_MAX) + 1);
   double s = d * (high - low);
   return low + s;
}
```



The random.cpp Implementation

```
/*
 * Implementation notes: randomChance
 * ----------------------------------
 * The code for randomChance calls randomReal(0, 1) and then checks
 * whether the result is less than the requested probability.
 */

bool randomChance(double p) {
   initRandomSeed();
   return randomReal(0, 1) < p;
}

bool randomBool() {
    return randomChance(0.5);
}
```



The `random.cpp` Implementation

```
/*
 * Implementation notes: setRandomSeed
 * -----------------------------------
 * The setRandomSeed function simply forwards its argument to
 * srand.  The call to initRandomSeed is required to set the
 * initialized flag.
 */

void setRandomSeed(int seed) {
   initRandomSeed();
   srand(seed);
}
```



The random.cpp Implementation

```
/*
 * Implementation notes: initRandomSeed
 * ------------------------------------
 * The initRandomSeed function declares a static variable that
 * keeps track of whether the seed has been initialized.  The
 * first time initRandomSeed is called, initialized is false,
 * so the seed is set to the current time.
 */

void initRandomSeed() {
   static bool initialized = false;
   if (!initialized) {
      srand(int(time(NULL)));
      initialized = true;
   }
}
```

The lifetime of **`static`** *variables begins the first time the program flow encounters the declaration and it ends at program termination. The compiler allocates only one copy of* **`initialized`**, *which is initialized exactly once, and then shared by all calls to* **`initRandomSeed`**. *This ensures that the initialization step must be performed once and only once.*